

FUNDAMENTAL INTUITIONS EXTENDED → SAT → H(S)H

The Glass Sausage Factory

An open, geometry-first research record. Begin with a picture, explore the public notebook, or follow the sources; the revisions and unfinished work stay in view.

- Browse the archive
- Conversation Viewer ↗
- Read Fundamental Intuitions Extended



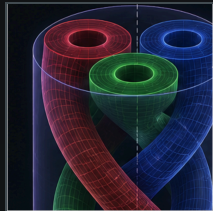
Time pushes obects to Earth. *Time* itself. I bet there's a way to call that entropy.

— Nathan Bellomy-McKnight

START ANYWHERE

Take a bite where you like.

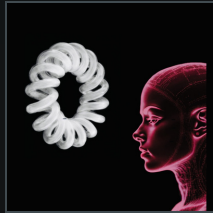
There is a quick route through the pictures, an essay, a spoken route, and the underlying record. They meet again further down.



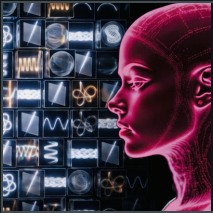
01 / SEE IT
Four moves behind the geometry
History, readout, recursion, inheritance.



02 / READ IT
Nathan's opening essay
The person and the method before the machinery.



03 / EXPLORE IT
The public notebook
PDFs, conversations, pictures, and reading rooms.



04 / TRACE IT
Sources and conversations
Documents, exact files, and the archive index.

- ON THE DESK**
- NOW
Two papers through review →
- IN THE NEWS
DESI and evolving dark energy →
- COMPARISON
Pulsar glitch dynamics →
- SITE
Latest updates →

FROM THE LIVE RECORD

The research desk

Current routing, selected science headlines, and worker submissions. Each item keeps its source and status.

- Large text
- What is moving
[Work feed](#) ↗
- In the news
[News context](#) ↓
- Factory floor
[Worker packets](#) ↗
- Listen

Routing Sep 24, 2026 · work snapshot Sep 20, 2026

ACTIVE ROUTING · SEP 24, 2026

Code Grurple — two-paper rapid peer-review/publication interlock

Operational priority · source linked

RAVEL / LIVE-THEORY SANDBOX

Finite-core H(s)H construction

Kerr/worldtube typing, cross-section and scale separation, null-helix/worldtube kinematics, force-interface questions, and related finite-core construction. Kerr and Kelvin remain live construction lines; live does not mean validated.

2 curated headlines · feed Sep 20, 2026

LAWRENCE BERKELEY NATIONAL LABORATORY / DESI COLLABORATION · MAR 19, 2025

DESI strengthens the case for looking beyond constant dark energy

comparison target

Related exploration ↗

LIGO-VIRGO-KAGRA COLLABORATION · JAN 14, 2025

The clearest gravitational-wave signal yet sharpens tests of the Kerr black-hole

5 worker submissions · candidate / ready

CALIPER / GPT-5.6 SOL · SEP 28, 2026 · READY

From Analogue to Calibration: SAT Geometry How-To

methodology / reproducibility workflow

[OBJECT OBJECT] · SEP 21, 2026 · CANDIDATE

Helical ruler: closure density and a single-carrier branch

Worker submission · editorial intake

FROM THE ARCHIVE · NATHAN

“T'd like to explore an idea I've had for a long time. I'll start by asking you to imagine a geometric space and ask you to answer some specific questions about that space.”

Nathan McKnight · source ↗

Site updates ↗

CONVERSATION FOCUS · FROM THE WORKING ARCHIVE

Forces Across Temporal Points

Read a preserved development conversation, follow the exchange turn by turn, or listen along with separate voices.

Started August 8, 2024

4 of 466

Conversation reading view

Listen along

Read conversation

First geometry conversation · 22 Mar 2024

Next conversation

Browse conversations

CHOOSE A READING DEPTH

Choose the depth that works for you. The two routes meet in the same source record.

Two clear ways in

PLAIN-LANGUAGE ROUTE

For curious readers

Start with pictures and the basic intuition: treat a particle not merely as a dot at an instant, but as part of a larger geometric history; ask how a lower-dimensional observer reads that structure; then follow what coiling, twisting, and braiding might organize.

1. The four generative intuitions — the conceptual kernel without formal prerequisites.

2. The visual argument — trajectory, readout, recursion, and braiding.

3. The 14-minute public debut ↗ — hear the idea in its earliest public form.

SCIENTIST-FACING ROUTE

For scientific readers

Begin with premise authority and current architecture, then move through translation and exact sources. Analogies, historical calculations, reconstructed derivations, computational experiments, and live H(s)H hypotheses retain different evidentiary labels.

1. BEDROCK premise register and current theory state.

2. SAT ↔ standard-physics translation desk and dated glossaries.

3. Formalization pipeline ↗, audits ↗, and claim/citation ledgers ↗.

4. [The Conversation Viewer ↗](#) — inspect how the ideas actually developed.

4. [Repository architecture ↗](#). Prior-art and citation work is presently being established as a source-separated public record.

START HERE · NATHAN’S STORY

Until very recently, I spent more time disavowing being a scientist than calling myself one.

Out of respect for the profession.

Not humility, exactly. I am not especially humble about the things I think I actually do well.

I am an editor.

asdfjkl;

Literally tattooed on my knuckles. Lowercase. Semicolon. Details matter.

I am a writer. When I am writing well, I am a damned fine writer, and I see no particular virtue in pretending otherwise.

I am a photographer. Not just in the sense that I own cameras. I have built working cameras out of Keebler tins and magnifying lenses and developed the photographs in my bathtub.

I am an educator.

And here, again, false modesty would just be another species of lying: I am a damned good educator.

• • •

Because somehow, accidentally, over the course of a lifetime of doing everything but being a physicist...

I taught myself to do a surprising amount of advanced physics in my fucking head.

Keep reading →

A LITTLE DEEPER / GENERATIVE
KERNEL

Fundamental Intuitions, extended

The cautious optimism here is specific. The foundational intuitions were designed as import-and-constraint rules: preserve successful physics, translate it into a small geometric vocabulary, expose conflicts, and let repeated cross-domain pressure select what survives. Their value is not that they guarantee the theory is right, but that there are strong rational reasons to expect them to generate useful structure and reveal where that structure fails.

World history

A particle-like object is modeled as an extended four-dimensional history, not only a succession of isolated local states.

Readout

What appears locally is related to an intersection or resolution of the larger geometric object.

Recursion

Coiling, curvature, torsion, bundling, and braiding repeat across scale and complexity.

Inheritance

Established physical success is a constraint to recover or explain—not permission to overwrite disagreement.

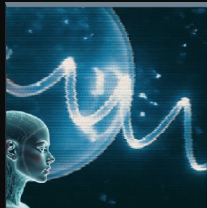
Read the presented edition

Open the original PDF

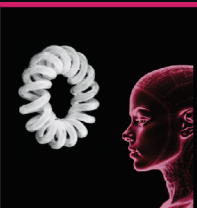
LISTEN EARLY / EXPLORE AT ANY
DEPTH

The spoken notebook

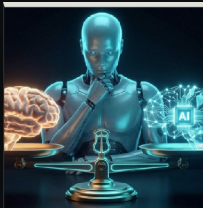
The same feed has carried several names and editorial eras—including *Debating AI on Consciousness and Reality*, *Debating A.I. on the Future of Physics*, and now *The New Physics*. The name changes are part of the record: an experimental public conversation widened into sustained theory development, criticism, archival release, and current H(s)H construction.



Consciousness and
dialogue era



Historical SAT visual
vocabulary



Human–AI method and
special-series artwork

ORIENTATION

How to read the project

The repository is an instrument, not a heap. Its navigation separates premises, source history, current synthesis, testing, and outside comparison so a polished sentence cannot quietly outrank its evidence. New readers can move through these six lanes.

Premises: BEDROCK

01

The living premise register. It marks material as foundational, working, tentative, contested, superseded, or direct-author—and states the active ancestry as FIE → SAT →

Excavation status is not theory status. A result not yet recovered, indexed, or typed by the current work crew has not thereby been shown absent from the corpus. The project’s discipline is to locate the source, preserve its date and wording, classify its authority, reconstruct the dependency chain, and only then test or translate it.



A New Way of Doing Physics

Sep 19 • The New Physics

Save on Spotify

31:25

2 FEB 2025 • 14:20 • SAT PUBLIC DEBUT What Is Thought Made Of? The first public Filament Unification introduction: the cleanest starting point for hearing the project before later vocabulary and machinery accumulated. Listen on Spotify ↗	5 MAR 2026 • 24:02 “Universal Indicatrix” Renormalization A concentrated entry into one of the major late-SAT solver and normalization programs. Listen on Spotify ↗	6 SEP 2026 • 48:21 Majorana Coupling & Coiled Cooper Pairing A route into the He-3, pairing, and contemporary SAT/H(s)H development lane. Listen on Spotify ↗
8 SEP 2026 • 37:51 Rotating Time Symmetry	19 SEP 2026 • 36:54 Theory of Everything Methodology	19 SEP 2026 • 31:26 A New Way of Doing Physics

Selected audio players



What Is Thought Made Of? [Filament Unification]

Feb 2 • The New Physics

Save on Spotify

14:20

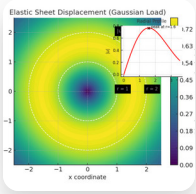


A New Way of Doing Physics

Sep 19 • The New Physics

Save on Spotify

31:25



Theory of Everything Methodology

Sep 19 • The New Physics

Save on Spotify

36:53



Rotating Time Symmetry

Sep 8 • The New Physics

Save on Spotify

37:51

DEVELOPMENT

A theory with a visible ancestry

The current model is neither a frozen 2025 theory nor a clean break from it. H(s)H attempts to retain SAT’s durable geometric grammar while rebuilding the mathematical object at finite-core worldtube resolution.

c. 1991

Recollected precursor

Early worldline/timesheet intuition. Treated as remembered provenance, not surviving documentary evidence.

[Full development timeline ↗](#) · [Audited status and history report ↗](#)

SOURCE MAP

Follow everything back.

Every synopsis on this page is secondary to the linked source surfaces. Repository lanes are intentionally separated so historical visibility, current authority, outside comparison, and operations are not confused.

H(s)H repository

H(s)H orientation / README

[Browse all 321 published episodes →](#)

The complete published title chronology is now indexed. Ninety-one episodes have exact, unambiguous transcript-file matches; concept, document, and remaining transcript joins stay open rather than being silently inferred.

[Search the episode index →](#) · [Browse podcast and transcript files ↗](#) · [Open the complete Spotify feed ↗](#)

● **WHAT NOW · ACTIVE PROJECT ROUTING**

Two papers through review.

An accelerated, source-linked pass is moving two papers from draft through independent review, revision, Nathan’s readiness check, and visibly SANDBOXED posting. Each step has its own status.

Independent review

Author revision

Nathan-ready check

SANDBOXED website posting

CURRENT ROUTING SNAPSHOT

Code Grurple

CHECK CURRENT STATE

Normal work is held at return cursors while the two paper streams move. A draft is not yet a reviewed site paper.

Source snapshot · 22 Sep 2026

PAPER A

Synthesis and provenance

Independent reviews have begun feeding an author revision. The [canonical packet ↗](#) carries the current draft and review state.

PAPER B

Boundary weak emission

A pre-review draft exists; authoring and source recovery continue. Follow the [paper packet ↗](#) for its actual stage.

STANDING RETURN CURSOR

* solver unification

The geometric solver work retains its source and next step while the paper interlock is active. It resumes on Nathan’s standdown, subject to what the paper work changes.

Routing changed since this snapshot · [check the current roster](#)

Public routing snapshot · source controls the current state

[Current roster ↗](#)[Signal queue ↗](#)

AFTER THE HEADLINES / LIVE PROGRAM

Current H(s)H and the audit lanes

The live program begins from the active ancestry documented in the repository: Fundamental Intuitions Extended → historical SAT corpus → reconstructed SAT backbone → live SAT solver machinery → SAT/H(s)H interface → finite-core worldtube construction.

BEDROCK premise state

Repository architecture

Library guide

History timeline

Source PDFs

The Fundamental Intuitions — Extended

Greenlight foundation · original PDF viewer.

FULL THEORY

SANDBOXED · historical formal SAT presentation.

https://glass-sausage-factory.nathanmcknight.chatgpt.site

6/18

CURRENT THEORY

H(s)H

Finite-core worldtubes, recursive hyperhelical carriers, bundle/braid geometry, boundary mechanics, and explicit readout.

[Repository ↗](#) · [State of theory ↗](#)

IN ESTABLISHMENT

Prior art & citation

A source-separated prior-art and citation program is presently being established. Public comparison records and citation surfaces will be linked when their provenance work is ready.

AUDIT

SAT footprint

Recovered claims are checked for originality, dependence, dimensions, operational observables, fit freedom, and whether apparent success was inserted by construction.

[Audits ↗](#) · [Ledgers ↗](#)

Current focus. Reconstruct the demonstrable SAT backbone while building and testing its finite-core H(s)H continuation. The conceptual bridge is substantially underway; the machinery- and equation-level bridge remains open.

RESULTS AND CONSTRAINTS

Results, constraints, and curious touchpoints.

Selected developments in physics that touch problems, mechanisms, or mathematical habits also present in SAT/H(s)H. Each item carries a typed relationship rather than being flattened into evidence for the project. Primary papers and official project pages come first; repository news dumps remain lead lists.

1 · Establish the result

What was measured, proved, or proposed—and with what

2 · Type the overlap

Shared topic, shared mathematics, distinctive mechanism, or genuinely

3 · Test chronology and contact

Public availability is a possible channel; diffusion requires evidence

uncertainty?

matching prediction?

beyond temporal possibility.

THEMATIC OVERLAP ONLY

DESI and evolving dark energy

DESI’s successive data releases make the expansion history of the universe an unusually sharp comparison surface. Reported preference for evolving dark energy remains a hint under active testing, not an established replacement for Λ CDM—and not evidence for SAT/H(s)H. A July 2026 Ly α analysis also moved toward Λ CDM, which is exactly why this lane follows the whole evidence stream.

[DESI DR2 guide ↗](#) [2026 milestone ↗](#) [July Ly \$\alpha\$ result ↗](#)

METHODOLOGICAL COMPARATOR

From hard spheres to hydrodynamics

A rigorous microscopic-to-macroscopic derivation is directly relevant to the program’s coarse-graining ambitions: it supplies an external standard for what “emergence from geometry/dynamics” must look like when made mathematically controlled. It is not a SAT derivation.

[Annals of Mathematics ↗](#) [Primary preprint ↗](#)

MECHANISM COMPARISON

Pulsar glitches and vortex dynamics

Contemporary glitch work provides concrete vortex-creep, pinning, superfluid and composite-vortex mechanisms against which SAT/H(s)H phase-snap language must be distinguished. Similar vocabulary alone is not a matching mechanism.

[Three glitches in a quiet pulsar ↗](#)

[Composite-vortex proposal ↗](#)

HOLOTYPE CONTEXT

Topological superfluid dynamics

Helium-3 remains an important experimental and conceptual comparison environment for the project. New work on chiral currents, surface waves, quasiparticles and confined phases should be read as external laboratory context—not retroactive support for an internal model.

[Physical Review Research ↗](#)

[Journal of Low Temperature Physics ↗](#)

Diffusion / convergence discipline. The default explanations for resemblance are independent convergence, shared background, or broad field-wide pressure. A public repository, podcast, or search result makes exposure possible; it does not show uptake. A serious diffusion claim needs chronology, distinctive shared content, a plausible discoverable channel, and preferably citation, contact, access evidence, or direct acknowledgment. Until then, the label here remains *thematic overlap only*.

Broader-science news archive

Source-rendered lead list · generated/copypaste material; use for wayfinding, not evidence.

Blind-chatter check archive

Source-rendered lead list · title and resemblance are not evidence of influence.

ETHOS

Glass,
including the
sausage.

SAT/H(s)H was developed largely through extended human–AI dialogue. The conversations are therefore part of the source record: not merely conclusions, but revisions, errors, discarded branches, moments of recognition, and the machinery that produced them. Public visibility is not offered as validation. It is offered so that provenance, dependence, and correction remain inspectable.

Explicit caution. SAT/H(s)H is an open research program under active development, not a peer-reviewed theory. Mathematical material ranges from Nathan-authored premises through AI-assisted derivations, computational experiments, repeated machine checks, and work awaiting independent specialist review. Polished presentation is not evidence of physical correctness.

“The archive is here to be read, tested, criticized, corrected, and improved.”

PRIMARY RECORD

Conversations

Browse the actual development dialogues through the public viewer, with their uncertainty and course corrections intact.

[Open Conversation Viewer ↗](#)

HISTORICAL ARCHIVE

GLASS

The public SAT archive preserves the 2023–25 theory record, older roots, papers, transcripts, calculations, and imagery.

[Open SAT archive ↗](#)

AUTHORITY CONTROL

BEDROCK

The living premise-state distinguishes foundational, working, tentative, contested, superseded, and direct-author material.

[Read BEDROCK ↗](#)

TRANSLATION DESK

Two vocabularies, one audit trail

SAT’s native terms are compact but can be opaque outside the archive. The translation desk makes them legible beside standard physics without declaring analogy to be equivalence. Each serious bridge should identify the source object, claimed function, limitation, proposed H(s)H counterpart, and present equivalence status.

Living Glossary v1

Search current meanings, historical usages, conventional comparisons, related terms, source trails, and status.

SAT → Standard, Part 1

Rendered in the reading room; original source remains one click away.

SAT → Standard, Part 2

Mappings, limitations, and unresolved formal obligations.

Historical SAT Glossary / Model Index v9

Dated terminology snapshot, explicitly distinguished from current authority.

SAT Concepts — May 2026

Raw GitHub text · dated conceptual snapshot for historical comparison.

Historical SAT Glossary

Raw GitHub text · track vocabulary across eras without projecting later usage backward.

Translation Compendium

Raw GitHub text · status must be checked at the claim level.

- filament
- timesheet
- θ₄
- readout
- worldtube
- Universal Indicatrix

PRESENTATION-READY SOURCE LAYER

A reading room, not a link wall.

The reader now surfaces PDF, plain-text, and Markdown material before repository machinery; keeps architecture, tooling, generated indexes, and workspaces one checkbox away; and gives plain-text sources a reading

Know what you are looking at

- PRESENTED EDITION
- PDF VIEWER
- SOURCE-RENDERED
- INDEX ROUTE
- MERSEARCH RESULTS

Presented editions are curated for reading. **PDF viewer** means the original file and

view beside their exact source text. Its new Index Desk connects the repositories’ different wayfinding systems without merging their roles.

Curated reading →

Open the Index Desk →

Browse both repositories →

pagination. **Source-rendered** improves display without conferring authority. **Index routes** are maps into sources. **Mersearch results** preserve source paths, query semantics, and provenance fields.

The source file—not the site rendering or index entry—controls exact quotation, pagination, authorship, date, and license.

IMAGE RECORD

The pictures are arguments and provenance.

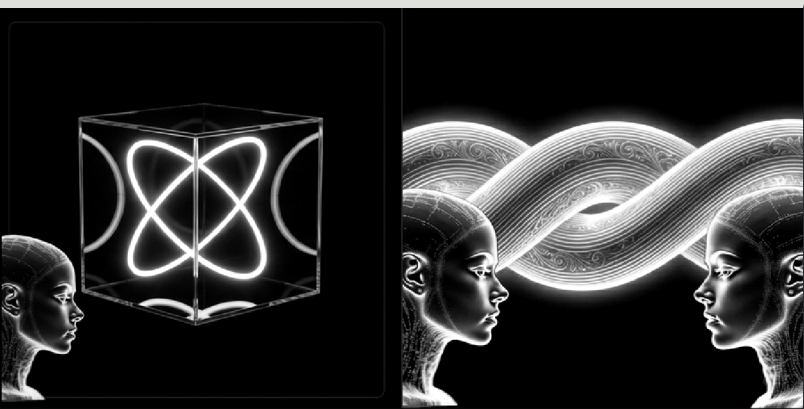
Visual material is presented as model construction, conceptual compression, or historical source—not as empirical demonstration. The image record connects early notebooks to later diagrams without claiming that every later interpretation was already present at the beginning.



HISTORICAL SAT / PUBLIC NOTEBOOK

Black, white, and signal magenta

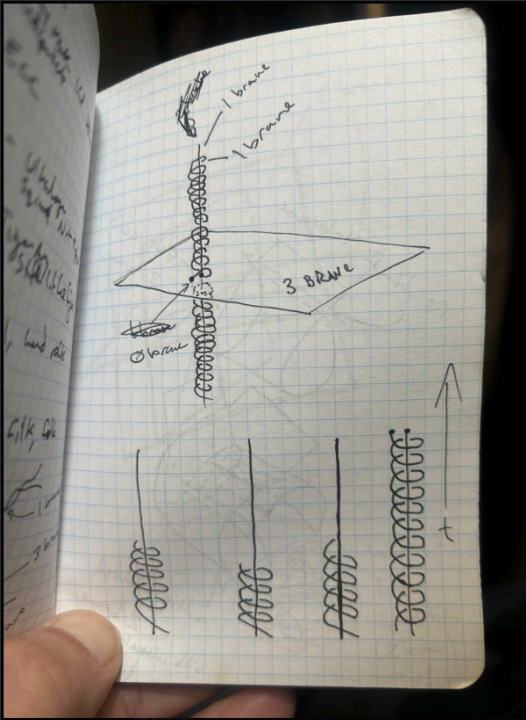
The observer figure, luminous constructions, and red–magenta accent identify SAT’s exploratory and public-facing visual record. These images orient a model; they are not data figures.



CURRENT H(S)H / TECHNICAL LAYER

Black, white, and measured gray

A quieter register for finite-core tubes, curvature, pitch, framed carriers, and current construction. Where a figure is computational rather than illustrative, its caption and source should say so.



Earliest surviving notebook material, 2003.



Trajectory / resolving-surface intersection.

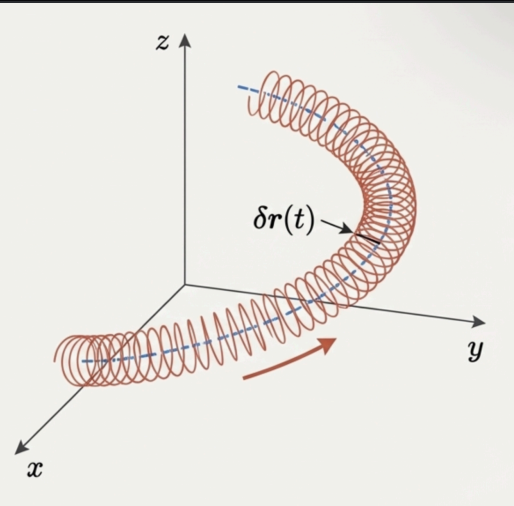
The Grammar of Interaction is Geometric Braiding

The complex rules of the strong force (QCD) are mapped as a direct geometric relationship. Three confined quark trajectories are treated as interbraided oscillations.

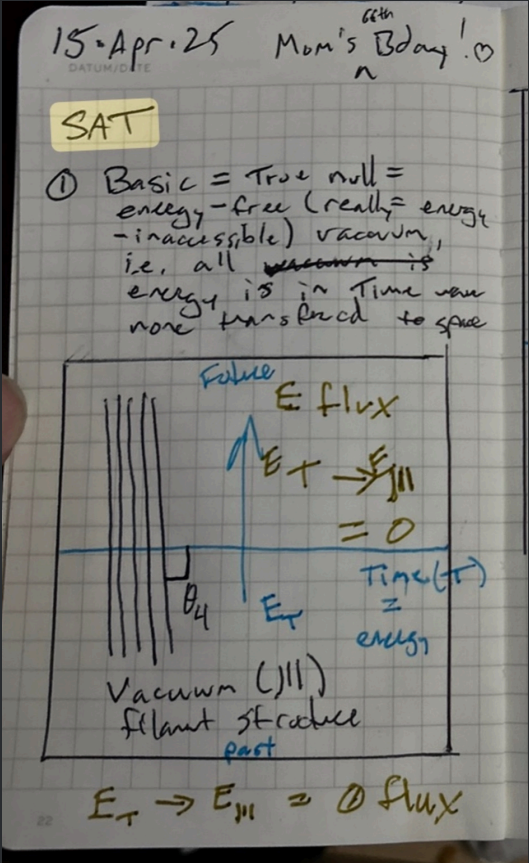
- **Color Charge:** Represented by the distinct colors of the coils.
- **Confinement:** The fact that quarks cannot be isolated is represented by the geometric necessity of their paths being braided together. The binding emerges from the geometry of intertwined oscillations, rather than an explicitly separate force field.



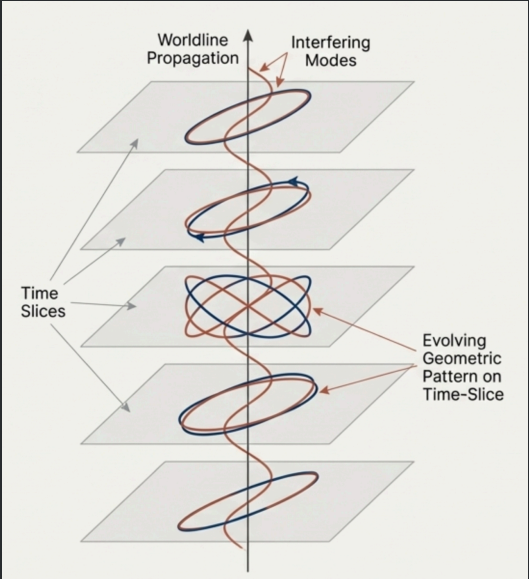
Braiding as an interaction grammar; a model mapping, not QCD derivation.



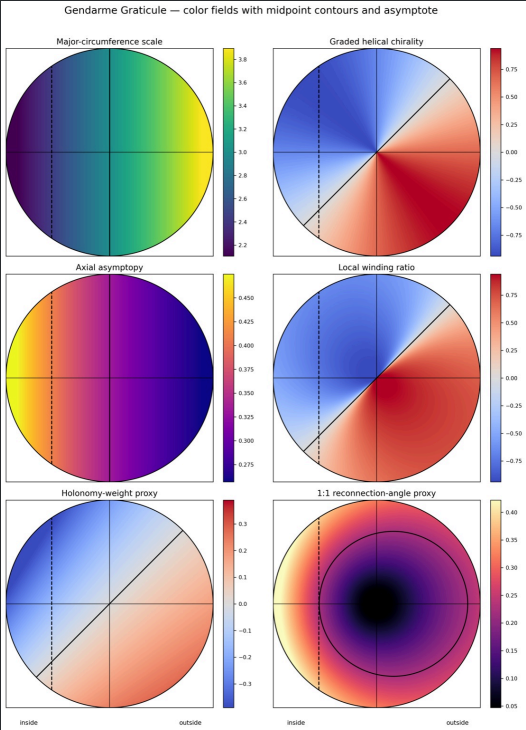
Recursive coiling on a curved carrier.



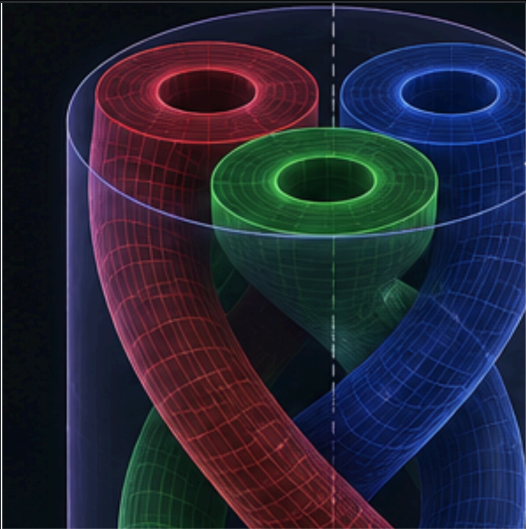
Explicit SAT notebook development, April 2025.



One continuous construction, successive observed patterns.



Gendarme Graticule computational field plot: technical figure, not decorative art. Interpretation remains tied to its generating definitions and code.



Concept rendering of a multi-tube bundle inside a containing boundary; illustrative, not an empirical reconstruction.



Exploratory field/path image. Retained as visual-development evidence, not a literal map of measured trajectories.

[Curated H\(s\)H visual collection ↗](#) · [Historical SAT visuals ↗](#) · [2003 sketches ↗](#)

PAPERS

Papers at work

Current manuscripts travel with version, review state, and source. The foundations and historical SAT papers have their own shelves.

SANDBOX MANUSCRIPT · R1

A direct search for weak secondary emission

The reviewed boundary-passage proposal is presented with its exact version and internal review trail.

Read the manuscript →

IN REVISION

Convergent geometric motifs

The comparative paper remains in revision. Its review record is available while the next manuscript version is prepared.

See its stage →

Open the full paper desk →

INTERACTIVE ARCHIVE · STILL IN DEVELOPMENT

See the conversations where the work happened.

Much of SAT/H(s)H was developed through extended human–AI dialogue. The Conversation Viewer opens that primary record directly: originating ideas, working definitions, wrong turns, corrections, rediscoveries, and the connective tissue behind many archive documents.

Browse and search the preserved conversations, replay their sequence, and jump to marked moments. New conversations will continue to be added as the corpus is processed, so the Viewer is both a historical window and a growing route into current reconstruction.

Primary conversational provenance

Growing corpus

Search + replay

Source before synopsis

Launch the Conversation Viewer



Viewer source and documentation



Development conversation files



PUBLIC-SITE ROADMAP

The original site-design conversation has been reconciled against the live site and its publication log. Completed foundations are not carried forward as open work;

What the public record needs next.

dependencies are not presented as deficits.

NOW LIVE

Living Glossary v1

Fifteen stable term addresses now separate current control, active scaffolds, historical translation, and recovery targets. [Open the glossary →](#)

NOW LIVE

Claims & Predictions Explorer v1

Six initial claim families now separate programme commitments, hypotheses, predictions, checks, dependencies, and source-recovery status. [Open the explorer →](#)

SOURCE-DEPENDENT

Mercer search feed

The reader already accepts Mercer result sets. Live archive results begin when a public query service or published result feed exists; the site does not pretend that endpoint is present now.

[Open the reconciled roadmap →](#)

THE EXPERIMENTALIST

About Nathan

Nathan Bellomy-McKnight is an independent researcher, writer, artist, podcaster, and longtime informal science communicator with a background in geological sciences and paleontology. His recurring method is experimentalist and cartographic: begin with established observations and successful theory, permit wide speculation, then cross-check, revise, and use geometry as a map of where to look—not as a substitute for evidence.

Science in public

Work in informal science education, including COSI and the American Museum of Natural History, informs a preference for explanations that can be inspected, pictured, challenged, and rebuilt.

Other work

The larger creative record includes fiction, visual art, music, consciousness and epistemology experiments, historical cookery, public conversations with AI, and the autobiographical “Great Moments in Personal Scientific History.”

Contact + identity

Nathan Bellomy-McKnight
[ORCID: 0009-0006-2462-0062 ↗](#)
scalarangulartheory@gmail.com

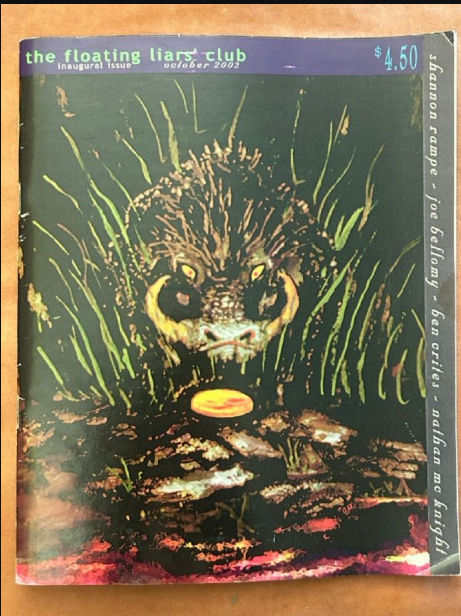
[See Nathan’s other projects →](#)

Working-paper note. A small number of papers also appear on Academia.edu. They should be read as draft-stage snapshots in a fast-moving development record; the dated repository sources and current status documents are the more reliable guide to what the program presently claims.

BEYOND THE RESEARCH

Nathan’s other projects.

The creative work has its own rooms. Follow the original sources and listen on the platforms where the releases live.



01 • LITERARY MAGAZINE

The Floating Liars’ Club

Three photographed issues from 2002–03. The separate site opens onto a page atlas and original facsimiles; the first reconstruction map grows as printed order, stories, and credits are checked.

[Enter the magazine →](#)

02 • MUSIC • DEDICATED SITE QUEUED

Music

A separate music home is planned. For now, these are Nathan’s supplied listening routes:

- 1. [First album on Spotify ↗](#)



2. [Artist page on Spotify ↗](#)

3. [Second album on Spotify ↗](#)

Archive-adjacent music artifacts, including n,n-dimethylmcfuqueroiserie and OS-00, are candidates for a future embedded or atmospheric layer after source and relationship review.

CITATION

Credit the source and the state.

Citations should identify Nathan Bellomy-McKnight, the SAT/H(s)H project, the exact document or artifact, its repository path, and an access date. Historical sources should be labeled historical; generated or reconstructed material should retain its authorship and vetting status.

[Citation ledger ↗](#)

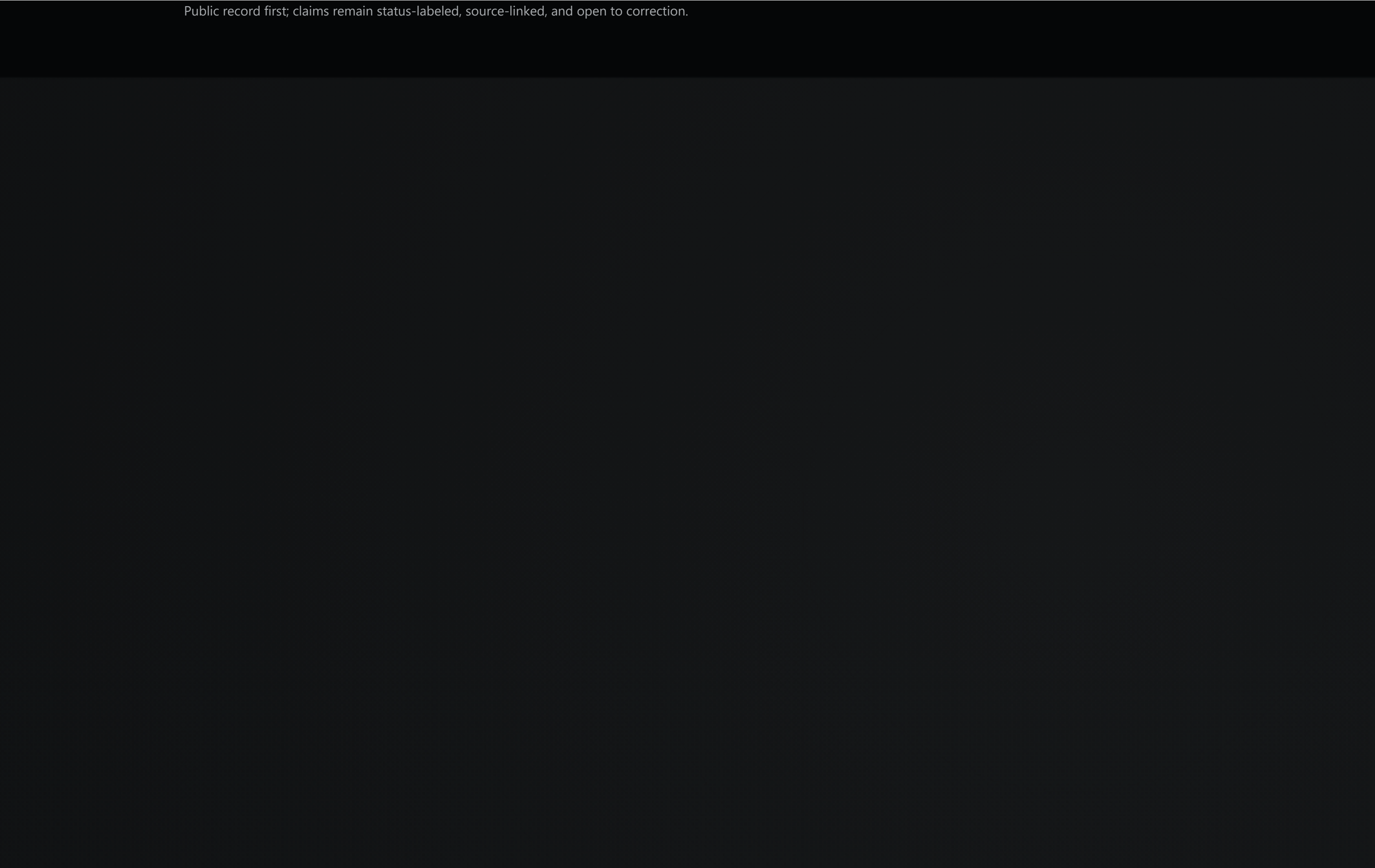
LICENSE

Open—with conditions made explicit.

The repositories carry their controlling license texts. Reuse should cite the project and preserve attribution, source distinctions, notices, and the difference between theory claims, archive materials, third-party material, and machine-generated contributions.

[H\(s\)H license ↗](#) · [SAT archive license ↗](#)

This site summarizes the repositories; it does not replace their license files or grant rights beyond them.



Public record first; claims remain status-labeled, source-linked, and open to correction.